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ENGINEERING



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Sreevidya Institute of Medical And Technical Sciences
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CSA15 Cloud Computing and Big Data Analytics

CO1 AT2 - Digital Concept Map and Service Model Matrix

Individual Student Submission Template

Student and Faculty Details

Student Name	SHAKTHI M	Register Number	192511197
Class / Section		Slot	C
Course Code	CSA1517	Course Title	CLOUD DATA AND BIG DATA ANALYSIS
Capstone Title	SleepScope: Student Sleep and Academic Stress Analytics App		
Submission Date	11/07/2026		
Faculty Name	Dr.Rajesh Babu	Employee ID	
Department	CSE		

Assessment Brief

Course Outcome	CO1 - Explain the evolution, architecture, characteristics, service models, and emerging paradigms of cloud computing.
Assessment Type	Individual concept mapping, service-model responsibility analysis, and capstone-cloud interpretation.
Assessment Focus	Each map must show key nodes, link labels, relationship meaning, and a logical hierarchy. The response must connect cloud fundamentals to the student's own perception of the capstone title.
Submission Mode	Editable DOCX + exported diagram evidence + GitHub repository link submitted in Google Classroom.
Permitted Tools	draw.io / diagrams.net, Lucidchart, Miro, Canva, PowerPoint, Google Drawings, or equivalent diagramming tool.
Individual Rule	Every student must submit their own concept map, matrix, capstone-cloud mapping, and reflection. The capstone title may be shared by a team, but the assessment answer must not be common.
Total Marks	100 marks

Code of Conduct and Student Assurance

Academic Integrity Statement

This assessment must be completed individually. Students may discuss the capstone domain with their team, but the concept map, service model matrix, explanations, GitHub evidence, and reflection submitted here must be individually prepared.

Student Assurance	I confirm that this submission represents my own understanding of CO1 concepts and my individual interpretation of the selected capstone title.
Tool Use	I have used digital tools only for diagram preparation, organization, and presentation. I have not copied another student's diagram, matrix, or explanation.
Evidence	I have uploaded the required files to my GitHub repository and submitted the repository link / evidence link through Google Classroom.
Student Signature	
Date	11/07/2026

Procedure for Preparing the Digital Concept Map

Use the following procedure to prepare the diagram and upload evidence. Students who already know another diagramming tool may use it, but the exported evidence and source file must still be submitted.

1. Open <https://app.diagrams.net/> or the desktop version of draw.io / diagrams.net.
2. Choose Device, Google Drive, or OneDrive as the save location according to availability.
3. Create a blank diagram and name the file as CSA15_CO1_AT2_RegisterNumber_Concept_Map.drawio.
4. Place the capstone title at the center or as a clearly visible anchor node.
5. Add major cloud nodes: cloud definition, cloud evolution, cloud architecture, service models, deployment models, virtualization, web services, SOAP, REST, APIs, elasticity, scalability, resource pooling, measured service, and emerging paradigms.
6. Connect nodes with labeled arrows such as uses, depends on, enables, scales through, communicates through, belongs to, supports, provides, consumes, or controls.
7. Use colors meaningfully: one color for service models, one for cloud characteristics, one for API/web-service concepts, and one for capstone-specific elements.
8. Export the final diagram as PNG or PDF. Keep the editable .drawio source file also.
9. Paste the exported diagram image or screenshot in this DOCX under Task 1.
10. Upload the DOCX, exported PNG/PDF, and .drawio file to GitHub, then submit the GitHub link in Google Classroom.

GitHub Submission Procedure

Step	Required Action	Expected Evidence
1	Create or use your individual GitHub account.	GitHub profile link.
2	Create a repository named CSA15-CO-Assessments-RegisterNumber or use the repository instructed by faculty.	Repository URL.
3	Create a folder named CO1_AT2 inside the repository.	Folder visible in repository.
4	Upload this completed DOCX, exported diagram PNG/PDF, and editable .drawio source file.	At least three evidence files uploaded.
5	Add a short README.md explaining the capstone title and files submitted.	README visible in the AT2 folder.
6	Submit the GitHub folder link in Google Classroom.	GCR submission with repository/folder link.

Assessment Questions and Student Response Sections

Important

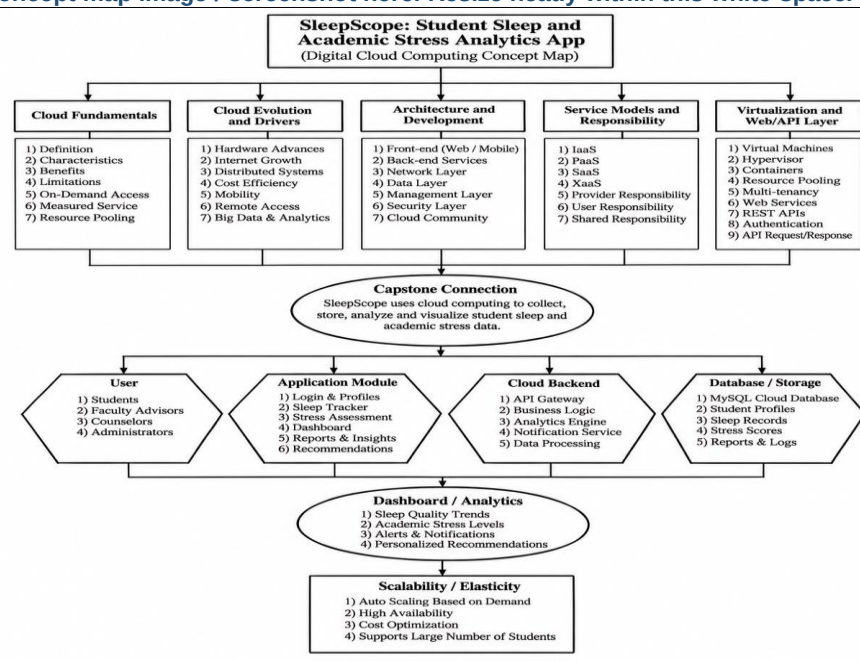
The capstone project may be developed by a team, but every response below must be written from the individual student's own perspective. Do not submit a common team answer.

Task 1 - Digital Cloud Concept Map

Create one integrated digital concept map for CO1. The map must show key nodes, link labels, relationship meaning, and a logical hierarchy. It must also connect the concepts to your selected capstone title from your individual perspective.

Mandatory Concept Group	Minimum Concepts / Nodes to Include
Cloud Fundamentals	Definition, characteristics, benefits, limitations, on-demand access, measured service.
Cloud Evolution and Drivers	Hardware evolution, internet software evolution, distributed systems, cost economics, mobility, remote access, big data growth.
Architecture and Deployment	Front-end, back-end, network layer, data center, management software, public/private/hybrid/community cloud.
Service Models and Responsibility	IaaS, PaaS, SaaS, XaaS, provider responsibility, user responsibility, shared responsibility.
Virtualization and Web/API Layer	Virtual machine, hypervisor, abstraction, resource pooling, multi-tenancy, web services, SOAP, REST, API request/response.
Capstone Connection	Your capstone users, app/module, cloud backend, database/storage, dashboard/analytics, scalability and elasticity idea.

Paste exported concept map image / screenshot here. Resize neatly within this white space.



Brief explanation of your concept map: explain the most important nodes, link labels, hierarchy, and capstone connection.

The concept map presents the major concepts of Cloud Computing in a logical hierarchy centered on the capstone project SleepScope: Student Sleep and Academic Stress Analytics App. The main node connects to Cloud Fundamentals, Cloud Evolution, Cloud Architecture, Service Models, Virtualization, Deployment Models, Web Services, REST APIs, Cloud Storage, Analytics, Security, Scalability, and Elasticity. Link labels such as includes, uses, stores, provides, analyzes, communicates through, and supports clearly describe the relationships among concepts. The capstone uses cloud computing to securely collect students' sleep duration, sleep quality, academic workload, and stress levels, store them in a cloud database, perform analytics, generate dashboards, and provide personalized recommendations. Cloud scalability enables the application to support thousands of students while maintaining high performance and availability.

Task 2 - Service Model Matrix

Complete the matrices by making technical decisions for your capstone project. Select only the rows that are relevant to your capstone design. If a row is not used, write a short technical reason instead of leaving it blank.

Your answer should show how responsibility shifts between cloud provider and student development team, and how the chosen services support the capstone workflow, API design, storage, security, scalability, and evidence submission.

Task 2A - Service Model Responsibility Matrix

Service Model	Capstone Role / Module	Provider Responsibility	Student Responsibility	Evidence to Submit
IaaS	Cloud server and database hosting	Provides virtual machines, storage, networking, security infrastructure	Configure server, database, APIs and deploy application	Cloud architecture diagram
PaaS	Application development and deployment	Manages operating system, runtime, middleware, platform	Develop, test and deploy SleepScope application	Deployment screenshot
SaaS	SleepScope Student Analytics Web Application	Maintains software availability, updates, backups	Manage student records, sleep analytics, stress reports and recommendations	Dashboard screenshot
XaaS / Specialized Service	Analytics API, Email Notification API, AI Recommendation Service	Provides API services, monitoring, scalability	Integrate APIs to generate alerts, reports and personalized recommendations	API integration screenshot

Task 2B - Capstone Cloud Service Decision Matrix

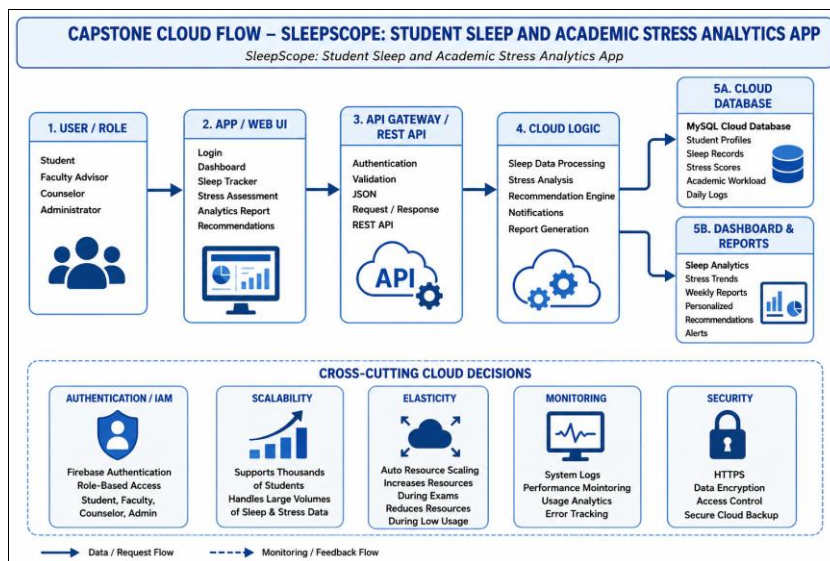
Cloud Layer / Function	Selected Tool or Service	Data / Resource Handled	API or Integration Path	Security / Scaling Note
Client access and UI	HTML, CSS, Bootstrap, JavaScript	Student dashboard, sleep tracker, stress questionnaire	Browser → REST API	HTTPS, responsive UI
API / backend logic	Python Flask / Django	Business logic, maintenance requests	REST API → Database	Auto scaling, secure APIs
Authentication and identity	Firebase Authentication	Student login and roles	Login API → Backend	Role-based Access Control
Database / structured records	MySQL Cloud Database	Student profiles, sleep records, stress data	Backend → MySQL	Encrypted data, regular backups
File / object storage	Firebase Storage / Cloud Storage	Reports, documents, images	App → Cloud Storage	Secure storage and scalable capacity
Monitoring / logs / usage metrics	Cloud Monitoring	System logs and performance	Server → Monitoring Service	Real-time monitoring and alerts
Analytics / reporting output	Power BI / Dashboard	Sleep analytics, stress trends	Database → Dashboard	Scalable reporting and data visualization

Technical justification: Explain why the selected cloud service pattern fits your capstone project's users, data, API flow, scalability, and security needs.

The selected cloud service architecture is ideal for the SleepScope: Student Sleep and Academic Stress Analytics App because students can securely access the application anytime using a web browser or mobile device. Sleep and stress information is stored in a cloud database and processed through REST APIs. Cloud services provide scalability for increasing numbers of users, while authentication, encryption, HTTPS communication, backups, and monitoring ensure security and reliability. Cloud analytics generate real-time dashboards that help students monitor their sleep habits and academic stress levels while enabling administrators to analyze anonymous trends.

Task 3 - Capstone Cloud Flow Mapping

Study the sample architecture-flow template below. Redraw it in draw.io / diagrams.net using your own capstone architecture. Replace every generic label with your actual user role, screen, API endpoint, backend logic, cloud storage, analytics output, security control, and scalability or elasticity decision.

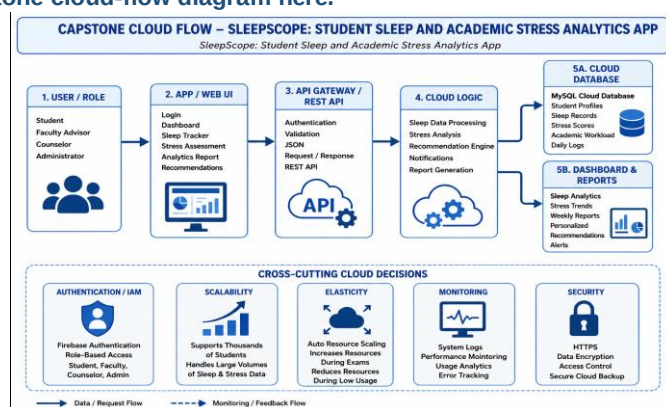


Student Drawing Requirement

Do not copy the sample as it is. Use it as a layout model, then create your own capstone-specific flow diagram with meaningful labels, arrows, data movement, API names, and cloud service choices.

Flow Component	Student Entry
User role and action	Students log in and record their daily sleep duration, sleep quality, study hours, assignments, and stress levels
Frontend / app screen	Login Page, Sleep Tracker, Stress Assessment Form, Dashboard, Analytics Report, Recommendations Page.
API endpoint and method	POST /login, POST /sleep, POST /stress, GET /dashboard, GET /analytics, PUT /profile.
Backend processing rule	The backend authenticates users, validates submitted sleep and stress information, stores data in the cloud database, analyzes patterns using analytics algorithms, calculates stress scores, generates recommendations, and updates dashboards.
Cloud database / storage operation	The MySQL Cloud Database stores Student Profiles, Sleep Duration, Sleep Quality, Academic Workload, Stress Scores, and Daily Logs. Cloud Storage stores exported reports and dashboard images.
Response, alert, report, or dashboard output	The dashboard displays Daily Sleep Hours, Sleep Quality Score, Academic Stress Level, Weekly Trends, Personalized Sleep Recommendations, Stress Alerts, and Analytics Charts.
Security, scalability, and elasticity labels added in diagram	Security: Firebase Authentication, HTTPS, Data Encryption, and Role-Based Access Control. Scalability: Supports thousands of students simultaneously. Elasticity: Automatically allocates additional cloud resources during peak usage, such as examination periods.

Paste your redrawn capstone cloud-flow diagram here.



Brief explanation: explain one complete request-response path from user action to cloud response.

When a student logs into SleepScope and records daily sleep information, the request is sent through a REST API to the cloud backend. The backend validates the user, processes the sleep and stress data, stores it securely in the MySQL cloud database, performs analytics, and generates personalized recommendations. The updated dashboard is then returned to the student, displaying sleep trends, stress levels, and health recommendations in real time.

Task 4 - Individual Reflection

Answer the following from your own perspective. Avoid team-common wording.

1. What is the strongest cloud concept represented in your capstone project?

The strongest cloud computing concept in my capstone project is **Software as a Service (SaaS)**. SleepScope is delivered as a web-based application that students can access from any location without installing software. Cloud services provide centralized storage, analytics, automatic updates, scalability, and secure access, making the application reliable and easy to use.

2. Where will elasticity or scalability become important in your capstone product?

Scalability and elasticity become important during examination periods when many students simultaneously access the application to record sleep information, complete stress assessments, and view analytics. The cloud automatically increases computing resources during peak usage and reduces them during normal periods, ensuring consistent performance while minimizing operational costs.

3. Which API or web-service idea is most relevant to your project and why?

The most relevant web-service technology for my project is the **REST API**. REST APIs enable seamless communication between the frontend, backend, cloud database, and analytics modules. They efficiently handle user authentication, sleep data submission, stress assessments, dashboard retrieval, and personalized recommendations while maintaining scalability, flexibility, and security.

4. What did you personally understand better after preparing this concept map and matrix?

After preparing the concept map and service model matrix, I gained a deeper understanding of how cloud computing technologies work together to build scalable applications. I learned how SaaS, PaaS, IaaS, REST APIs, virtualization, cloud databases, analytics, security, scalability, and elasticity collectively support the development of the **SleepScope: Student Sleep and Academic Stress Analytics App**. This exercise improved my understanding of designing secure, efficient, and cloud-enabled healthcare and analytics solutions.

Evaluation Rubric - 100 Marks

Criteria	Marks	Expected Standard
Concept identification	20	Major CO1 concepts are accurately identified: cloud fundamentals, characteristics, evolution, service models, deployment models, virtualization, web services, SOAP/REST, APIs, elasticity, and scalability.
Relationship mapping	20	Links between concepts are accurate, meaningful, and labeled. The map should explain how concepts depend on, enable, consume, provide, scale, or control one another.
Hierarchy and organization	15	Concepts are arranged in a logical hierarchy with clear grouping, readable flow, and no disconnected keyword clusters.
Technical relevance	15	The map strongly reflects cloud course content and avoids decorative or
		generic non-technical content.

Service model matrix	15	IaaS, PaaS, SaaS, and XaaS are compared with provider responsibility, user responsibility, examples, capstone usage, and risks or limitations.
Capstone alignment and individual reflection	10	The student's own capstone interpretation is visible in the concept map, flow mapping, justification, and reflection. Team-common answers are avoided.
Visual clarity and digital evidence	5	Diagram is neat, readable, exported properly, and supported with GitHub evidence including source and exported files.

Faculty Evaluation Record

Evaluator Name		Marks Awarded	
Observation		Signature	